T|W|I
Thurgauer
Wirtschaftsinstitut
an der Universität Konstanz

When personalization deters delegation to algorithms: Experimental evidence

Lara Suraci & Irenaeus Wolff

— 23rd June, 2026, Symposium on 'AI and the Future of Work', Berlin —

Background

- Huge & unconnected literature on algorithm/automation/AI/... reliance
- Most of the literature: setups with clearly identifiable mistakes
- But in many real-life settings, preferences determine what is a mistake
- E.g., robo-advisors: maximal payoff *vs* best fit to risk-parameters

Research Question

TWI
Thurgauer Wirtschaftsinstitut
an der Universität Konstanz

- Does algorithm reliance differ for **objective optimization *vs.* preference-based personalisation** in a preferential-choice environment?
  (i) ...at first contact?
  (ii) ...overall?

Objective vs Personalised Algorithms

- What do we mean by **objective optimization**?
  - an algorithm that chooses some 'objectively ideal' option independently of a subject's preferences (here: **highest EV**)
- What do we mean by **personalised**?
  - an algorithm that measures a subject's preferences and chooses accordingly (here: **implementation of revealed preferences**)

Objective vs Personalised Algorithms

- What do we mean by **objective optimization**?
  - an algorithm that chooses some 'objectively ideal' option independently of a subject's preferences (here: **highest EV**)
- What do we mean by **personalised**?
  - an algorithm that measures a subject's preferences and chooses accordingly (here: **implementation of revealed preferences**)
- In the experiment, we describe both algorithms in intuitive terms:

EV: *'The algorithm works out how much each option would yield on average and selects the option for which this is highest.'*

Prefs: *'The algorithm analyses your choices from Part 1 and selects the best option for you in light of your Part-1 choices.'*

Relevance?

Well-known (Swiss) robo-advisors(' marketing strategies)

- "We rely on a low-cost, passive investment strategy and have selected a total of 9 equity, bond and real estate ETFs from over 1,500 ETFs (exchange-traded funds) and put them together in such a way that they achieve an optimal risk/return ratio." (findependent.ch; for GER, see e.g., whitebox.eu)
- "Chat with Selma: Selma looks at your financial life and learns how you should structure your finances. Get your plan: Selma creates a personalized investment plan that fits your life. Sit back and relax! Selma takes care of your money from now on and does all the buying and selling for you." (selma.com; for GER, see e.g., quirion.de)

Experimental Design

Thurgauer Wirtschaftsinstitut an der Universität Konstanz

Part 1 – Elicitation of preferences

- only two, 'well-presented' lotteries, unlimited time
- "hypothetical, but the choices may be meaningful for Part 2"

Part 2 – Lottery Selection Task

- could stand for an investment task
- complexity (number of options, outcomes under different SOTW)
- time pressure (think: relative to the number of options)
- each Part-2 task built from 2 anchor options (→ Part 1) + 6 decoys

Tasks

	State 1	State 2	State 3	State 4	State 5	State 6	State 7	State 8	State 9	State 10
Option 1	225	225	225	225	225	225	225	205	0	0
Option 2	525	525	525	525	525	0	0	0	0	0

State:	1	2	3	4	5	6	7	8	9	10
Option 1	225	225	0	225	0	225	225	225	225	0
Option 2	525	0	505	0	525	0	0	0	525	505
Option 3	525	0	525	0	525	0	0	0	525	525
Option 4	225	225	0	225	205	205	205	225	225	0
Option 5	225	205	0	205	205	225	225	205	225	0
Option 6	225	225	0	225	205	225	225	225	225	0
Option 7	525	0	525	0	525	0	0	0	225	525
Option 8	505	0	525	0	505	0	0	0	505	525

- For further distraction: 4 filler tasks in between
- 8 rounds: 5’ pre-view — 5’ algorithm choice—8’ decision/waiting time

Tasks

	State 1	State 2	State 3	State 4	State 5	State 6	State 7	State 8	State 9	State 10
Option 1	225	225	225	225	225	225	225	205	0	0
Option 2	525	525	525	525	525	0	0	0	0	0

State:	1	2	3	4	5	6	7	8	9	10
Option 1	225	225	0	225	0	225	225	225	225	0
Option 2	525	0	505	0	525	0	0	0	525	505
Option 3	525	0	525	0	525	0	0	0	525	525
Option 4	225	225	0	225	205	205	205	225	225	0
Option 5	225	205	0	205	205	225	225	205	225	0
Option 6	225	225	0	225	205	225	225	225	225	0
Option 7	525	0	525	0	525	0	0	0	225	525
Option 8	505	0	525	0	505	0	0	0	505	525

- For further distraction: 4 filler tasks in between
- 8 rounds: 5’ pre-view — 5’ algorithm choice—8’ decision/waiting time

Tasks

Thurgauer Wirtschaftsinstitut an der Universität Konstanz

	State 1	State 2	State 3	State 4	State 5	State 6	State 7	State 8	State 9	State 10
Option 1	225	225	225	225	225	225	225	205	0	0
Option 2	525	525	525	525	525	0	0	0	0	0

State:	1	2	3	4	5	6	7	8	9	10
Option 1	225	225	0	225	0	225	225	225	225	0
Option 2	525	0	505	0	525	0	0	0	525	505
Option 3	525	0	525	0	525	0	0	0	525	525
Option 4	225	225	0	225	205	205	205	225	225	0
Option 5	225	205	0	205	205	225	225	205	225	0
Option 6	225	225	0	225	205	225	225	225	225	0
Option 7	525	0	525	0	525	0	0	0	225	525
Option 8	505	0	525	0	505	0	0	0	505	525

- For further distraction: 4 filler tasks in between
- 8 rounds: 5' pre-view — 5' algorithm choice—8' decision/waiting time

Sample, Incentives & Procedure

- Recruitment: Prolific UK, online (z-Tree); preregistered target 120/treatment before exclusions.
- Exclusions (preregistered): bot/slider checks, comprehension after 2nd attempt, self-reported AI use, super-fast log-times, incomplete data.
- Final sample: **N = 233** (EV: $n = 118$, Prefs: $n = 115$).
- Incentives: GBP 2.00 base + one randomly drawn Part-2 round (1 penny / point); up to GBP 8.90 bonus. Mean earning GBP 4.14, median duration ~16 min.

Preregistered Hypothesis & Analysis Plan

- **H1.** People are more likely to delegate financial decisions to an **EV-maximising** algorithm than to a **personalised** algorithm that incorporates their (revealed) preferences.
- Primary tests (preregistered):
  1. One-sided **Boschloo** exact test on **Round-1** delegation (first contact)
  2. One-sided **WMW** test on participant-level mean delegation across Part 2
- Preregistered LPM: decision-set FE, participant-clustered SE, treatment × period interaction, share of prior delegated rounds with zero payoff.
- Exploratory: categorical *never-delegation*, win-stay/lose-shift, ex-post reason items.

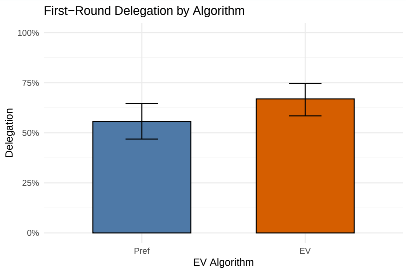
Main Results: First Contact

- Benchmark: only ~34–35% of *self-chosen* options were non-dominated; delegated rounds paid on average ~25% more than self-decided ones ⇒ delegation is the instrumentally rational thing to do.
- Round 1: **66.9% (EV)** vs. **55.8% (Prefs)**.
- Boschloo exact test, one-sided: $p = 0.0413$ ⇒ **H1 supported at first contact.**
- Treatment gap concentrated in the first decision.

Main Results: First Contact

Thurgauer
Wirtschaftsinstitut
an der Universität Konstanz

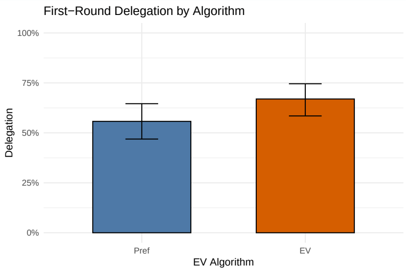
- Benchmark: only ~34–35% of *self-chosen* options were non-dominated; delegated rounds paid on average ~25% more than self-decided ones ⇒ delegation is the instrumentally rational thing to do.
- Round 1: **66.9% (EV)** vs. **55.8% (Prefs)**.
- Boschloo exact test, one-sided: $p = 0.0413$ ⇒ **H1 supported at first contact.**
- Treatment gap concentrated in the first decision.



Main Results: First Contact

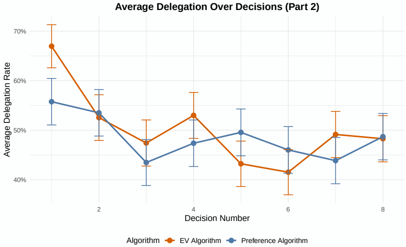
Thurgauer Wirtschaftsinstitut an der Universität Konstanz

- Benchmark: only ~34–35% of *self-chosen* options were non-dominated; delegated rounds paid on average ~25% more than self-decided ones ⇒ delegation is the instrumentally rational thing to do.
- Round 1: **66.9% (EV)** vs. **55.8% (Prefs)**.
- Boschloo exact test, one-sided: $p = 0.0413$ ⇒ **H1 supported at first contact.**
- Treatment gap concentrated in the first decision.



Main Results: Repeated Decisions

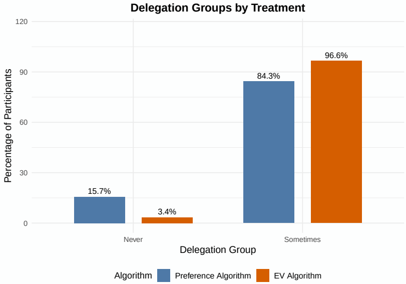
- Average delegation: no significant treatment difference overall (WMW one-sided $p = 0.2425$).
- Preregistered LPM (decision-set FE, clustered SE):
  - EV main effect: $\beta = -0.049$, $p = 0.56$
  - EV × period: $\beta = 0.002$, $p = 0.87$
  - Share of prior *zero*-payoff delegated rounds: $\beta = -0.385$, $p < 0.001$
- Convergence after Round 1 driven more by a decline in EV than by a rise in Prefs.
- But: first-contact decisions can determine whether users *come back at all.*



Categorical Non-Delegation (not preregistered)

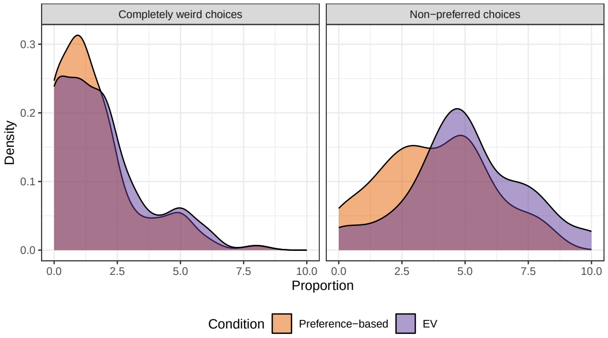
TWI
Thurgauer Wirtschaftsinstitut
an der Universität Konstanz

- Never-delegators:
  - **15.7%** in Prefs (18/115)
  - **3.4%** in EV (4/118)
- Always-delegators show the opposite, weaker pattern (13.6% Prefs vs. 19.1% EV).
- Never-delegators do *not* differ in age, income, education, gender, time spent, financial risk-taking, AI usage, or reliability.
- Within Prefs: never-delegators report (weakly) *higher* satisfaction with Part-1 choices ($\Rightarrow$ not driven by dislike of the elicited profile).
- Replication in earlier (PhD-thesis) study: 32/189 vs. 17/187, Boschloo $p = 0.0135$.



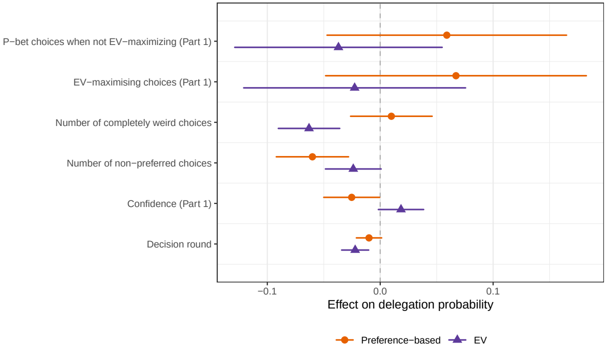
Mechanisms: *What* deters delegation differs by algorithm

- LPM of individual delegation on *expected non-preferred* and *expected "completely weird"* choices, × treatment, controlling for round, Part-1 confidence, risk preferences, task FE:



Mechanisms: *What* deters delegation differs by algorithm

- LPM of individual delegation on *expected non-preferred* and *expected "completely weird"* choices, × treatment, controlling for round, Part-1 confidence, risk preferences, task FE:

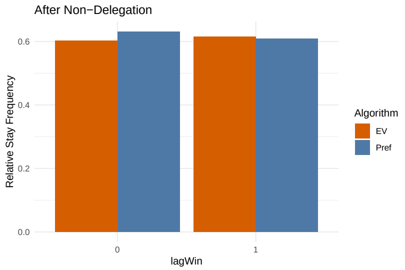


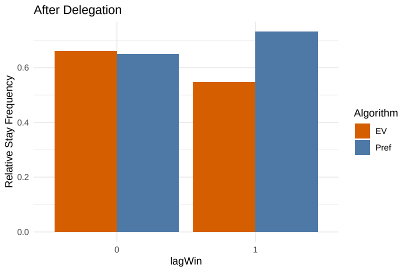
95% confidence intervals. Linear probability model with task fixed effects and part
EV−condition effects are main effect + interaction term.

Mechanisms: Self-reported reasons

- Differences run more by *delegator type* than by treatment.
- Delegators (vs. never-delegators) more often cite performance motives (70.1% vs. 36.4%: "computer would choose better").
- Never-delegators more often cite distrust of the algorithm and responsibility for the choice (63.6% vs. 38.9%).
- Crucially: they want responsibility for the choice, *not* for the outcome $\Rightarrow$ this is a positive valuation of **choice authority**, not blame-avoidance.
- Items on being algorithmically analysed and on opacity ("don't know how it chooses") are rarely selected $\Rightarrow$ not a transparency story.
- Consistent with *control / decision-autonomy* as a driver of acceptance (Schaap et al., 2024; Ivanova-Stenzel & Tolksdorf, 2025).

Dynamics: win-stay/lose-shift





Summary & Conclusion

TWI Thurgauer Wirtschaftsinstitut an der Universität Konstanz

1. Preregistered behavioural comparison of *objective* vs *personalised* decision rules ⇒ personalisation does *not* increase delegation.
2. The effect is concentrated at first contact (Round 1: 66.9% EV vs 55.8% Prefs) — and first contact often decides whether users return at all.
3. Categorical non-delegation is a distinct uptake barrier under personalisation (15.7% vs 3.4%), *not* explained by demographics, risk attitudes, AI exposure, or dissatisfaction with the elicited profile.

Thank you!

Thank you — and looking forward to your questions.

Irenaeus Wolff · wolff@twi-kreuzlingen.ch

Lara Suraci · DG Cities

Preregistration: https://aspredicted.org/7un5qf.pdf